
Calculators and communication devices are not allowed.

Answer the following questions.

1. (a) (10 pts.) Let A be an $n \times n$ matrix. Prove that, if A is invertible, then the linear system $AX = B$ has exactly one solution for each $n \times 1$ matrix B .
- (b) Let $X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$.
 - (i) (8 pts.) Find a symmetric matrix S and a skew-symmetric matrix K such that $X = S + K$.
 - (ii) (7 pts.) Find X^{-1} and then, find B , if $BX = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$.
2. (a) Let $A = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 & 1 \\ -1 & -2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$.
 - (i) (10 pts.) Find A^{-1} , if it exists.
 - (ii) (5 pts.) Solve, if possible, the homogeneous linear system $(A - B)X = O$.
- (b) (10 pts.) Show that a linear system can have zero, one or infinitely many solutions.
3. (a) (10 pts.) Let A and B be $n \times n$ matrices such that $BA = I_n$. Show that $B = A^{-1}$.
- (b) (15 pts.) Let A be an invertible symmetric matrix. Show that A^{-k} is symmetric, for any positive integer k .
4. (a) (15 pts.) Find all values of a, b, c , and d for which A is skew-symmetric

$$A = \begin{bmatrix} 0 & a - 3b + c & a - 4b + c \\ -2 & 0 & 5 \\ -3 & a - 3b + 2c & d \end{bmatrix}.$$

1. In each of the following questions, select the correct answer. (2 pts. each)
- (I) Let A and B be 3×3 matrices such that $\text{tr}(A) - \text{tr}(B) = 5$, then
 - (A) $A - B = 5I_3$
 - (B) $\text{tr}(A - B) = \text{tr}(5I_3)$
 - (C) $\frac{3}{5}\text{tr}(A - B) = \text{tr}(I_3)$
 - (D) None of the above
- (II) If the linear systems $AX = B$ and $CX = D$ are consistent and have the same solution X_0 , then
 - (A) $B = D$, if $A = 2C$
 - (B) A and C are invertible
 - (C) $A = C$, if $B = D$
 - (D) None of the above
- (III) Let A and B be invertible matrices, then
 - (A) $\text{tr}(AB) \neq 0$
 - (B) $\text{tr}(A) - \text{tr}(B) = 0$
 - (C) $\text{tr}(A) = 0$
 - (D) None of the above
- (IV) Let X and Y be $n \times n$ diagonal matrices. Then
 - (A) X is nonsingular
 - (B) $(XY)^2 = X^2Y^2$
 - (C) X and Y are nonsingular
 - (D) None of the above
- (V) Let X and Y be $n \times n$ invertible matrices. Then
 - (A) $X^2(Y^3)^T$ is invertible
 - (B) $X^2 - Y^3$ is invertible
 - (C) $X + Y - 3XY$ is invertible
 - (D) All of the above

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Answer the following questions.

1. (a) (10 pts.) Let A be an $n \times n$ invertible matrix. Then A^{-1} exists and $A^{-1}(AX) = (A^{-1}A)X = I_n X = X = A^{-1}B \implies X = A^{-1}B$ is a solution of the system. Suppose that X_0 is any solution of the system $AX = B$, then $AX_0 = B$. thus, $A^{-1}(AX_0) = A^{-1}B$, and then $X_0 = A^{-1}B$.
- (b) $X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \implies X^T = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$
 - (i) (8 pts.) $S = \frac{1}{2}(X + X^T) = \begin{bmatrix} 1 & \frac{3}{2} \\ \frac{3}{2} & 3 \end{bmatrix}$ & $K = \frac{1}{2}(X - X^T) = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$.
 - (ii) (7 pts.) $X^{-1} = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$. $BX = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \implies B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} X^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.
2. (a) (i) (10 pts.) $[A|I_3] = \left[\begin{array}{ccc|ccc} 0 & -1 & 1 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -4 & -1 & 2 \\ 0 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \implies$
 $A^{-1} = \begin{bmatrix} -4 & -1 & 2 \\ -2 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$.
(ii) (5 pts.) $A - B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. The augmented matrix of $(A - B)X = O$ is
 $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \equiv \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$. The solution is $x = r, y = 0, z = t$,
where $r, t \in \mathbb{R}$.
(b) (10 pts.) See book.
3. (a) (10 pts.) (I) A is invertible:
Consider the system $AX = O \implies B(AX) = BO = O \implies (BA)X = O \implies I_n X = O \implies X = O$. Thus, $AX = O$ has only the trivial solution. Therefore, A is invertible.
(II) $B = A^{-1}$:
Since A is invertible, then A^{-1} exists. Hence, $A^{-1}(AB) = A^{-1}I \implies (A^{-1}A)B = A^{-1} \implies IB = A^{-1} \implies B = A^{-1}$.
- (b) (15 pts.) $(A^{-k})^T = ((A^{-1})^k)^T = \left(\underbrace{A^{-1}A^{-1}\dots A^{-1}}_{k\text{-times}} \right)^T = \underbrace{(A^{-1})^T (A^{-1})^T \dots (A^{-1})^T}_{k\text{-times}} =$
 $\underbrace{(A^T)^{-1} (A^T)^{-1} \dots (A^T)^{-1}}_{k\text{-times}} = ((A^T)^{-1})^k = (A^{-1})^k = A^{-k}$.
Or: $(A^{-k})^T = ((A^{-1})^k)^T = ((A^k)^{-1})^T = ((A^k)^T)^{-1} = \left(\left(\underbrace{AA\dots A}_{k\text{-times}} \right)^T \right)^{-1} =$

4. (a) (15 pts.) Since A is skew-symmetric, then $a-3b+c=2$, $a-4b+c=3$, $a-3b+2c=-5$ and $d=0$. Thus,

$$\left[\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 1 & -4 & 1 & 3 \\ 1 & -3 & 2 & -5 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -7 \end{array} \right] \implies a=6, b=-1, c=-7 \text{ and } d=0.$$

- (b) (I) Let A and B be 3×3 matrices such that $\text{tr}(A) - \text{tr}(B) = 5$, then
 (C) $\frac{3}{5}\text{tr}(A - B) = \text{tr}(I_3)$
 (II) If the linear systems $AX = B$ and $CX = D$ have the same solution X_0 , then
 (D) None of the above
 (III) Let A and B be invertible matrices, then
 (D) None of the above
 (IV) Let X and Y be $n \times n$ diagonal matrices. Then
 (B) $(XY)^2 = X^2Y^2$
 (V) Let X and Y be $n \times n$ invertible matrices. Then
 (A) $X^2(Y^3)^T$ is invertible